# After Cardiac Surgery

CSC focus: bleeding, tamponade, shock states, POAF, pacing and CSU-ALS



## FIRST HOURS AND BLEEDING

- BP: MAP about 60 to 90; avoid hypertension (worsens bleeding)
- Labs: ABG, Hgb, K, Mg, ionized Ca, coags; glucose 180 mg/dL or less (insulin drip)
- Rewarm to 36 °C or more; expect vasodilation; treat shivering (raises O2 demand)

*Bleeding with instability also qualifies; follow your institution's protocols.*

| CHEST TUBE OUTPUT | CLASSIC RETURN-TO-OR TRIGGER |
|---|---|
| **1st hour** | Over 500 mL |
| **Each of first 2 h** | Over 400 mL/h |
| **Each of first 3 h** | Over 300 mL/h |
| **First 4 h total** | Over 1,000 mL |

*Best guided by TEG/ROTEM; also correct pH and ionized Ca.*

| PROBLEM | TYPICAL FIX |
|---|---|
| **Residual heparin or rebound (ACT up)** | Protamine |
| **Fibrinogen under 150 mg/dL, oozing** | Cryo or fibrinogen concentrate |
| **Factor deficiency** | PCC or plasma |
| **Platelet deficit or dysfunction** | Platelets (DDAVP if dysfunction) |
| **Fibrinolysis** | Tranexamic acid |

Protamine: 1 mg reverses at least 100 units heparin; EACTS 2024 advises under 1:1 of the initial heparin dose. Very slow IV: 50 mg max over 10 min. Reactions: hypotension, anaphylaxis, pulmonary hypertensive crisis (NPH insulin, fish allergy, vasectomy, prior protamine). Heparin rebound: 30 min to 18 h post-op.

**SAFETY** Heavy bleeding with normal coags suggests a surgical source. Delayed re-exploration worsens outcomes; escalate early.

## TAMPONADE, LOW OUTPUT OR VASOPLEGIA?

| PROBLEM | FILLING PRESSURES | CI | SVR |
|---|---|---|---|
| **Bleeding** | Low | Low | High |
| **Tamponade** | High, equalizing | Low | High |
| **LV failure** | PAOP high | Low | High |
| **RV failure** | CVP over PAOP | Low | High |
| **Vasoplegia** | Low to normal | Normal or high | Low |

**EXAM TIP** Exam favorite: drainage suddenly stops (clotted tubes) while CVP, PA diastolic and PAOP rise and equalize and pulse pressure narrows. Think tamponade; call the surgeon for re-exploration.

Often a localized clot: Beck's triad may be absent; TEE beats TTE.

Low output (LCOS): CI under about 2.0 to 2.2 with poor perfusion despite filling; stunning peaks about 4 to 8 h post-op. Optimize preload, rate, AV synchrony and electrolytes; add an inotrope. Escalate early to IABP, Impella or VA-ECMO, before lactate reaches 4.

Vasoplegia: MAP under 65, CI over 2.2, SVR under 800 (risks: long bypass, preop ACE inhibitors). Norepinephrine plus vasopressin; if refractory, methylene blue 1 to 2 mg/kg (serotonin syndrome with SSRIs, falsely low SpO2, G6PD hemolysis) or hydroxocobalamin 5 g (red urine, lab interference, CRRT blood-leak alarms). Follow your institution's protocols.

## POST-OP AF AND EPICARDIAL PACING

POAF: 30 to 50%, more after valve surgery; peaks around day 2. STS 2026 prevention: oral amiodarone (Class I), beta-blockers or posterior pericardiotomy (IIa). Unstable: cardioversion. Stable: rate or rhythm control, replace K and Mg; anticoagulate by stroke risk once bleeding settles.

*Atrial wires usually exit right, ventricular left; confirm labels.*

| MODE | TYPICAL USE |
|---|---|
| **AAI** | Sinus brady, AV conduction intact |
| **DDD** | AV block; keeps atrial kick |
| **VVI** | Backup, or slow AF |




Failure to sense: spikes ignore intrinsic beats; one on a T wave can trigger VF.

**SAFETY** Wear gloves; insulate exposed wire ends (microshock). After wire removal, watch for tamponade.

## CABG VS VALVE, AND THE STERNUM

| SURGERY | WATCH FOR |
|---|---|
| **CABG** | Ischemia: ST changes, VT/VF; aspirin within 6 h |
| **Radial graft** | Spasm: CCB often for 1 year |
| **AVR** | Heart block; keep wires connected |
| **Mechanical valve** | Lifelong warfarin: INR 2.5 AVR (3.0 if risk factors), 3.0 MVR; no DOACs |
| **Bioprosthetic** | Aspirin; sometimes warfarin 3 to 6 months |

Sternum: Keep Your Move in the Tube (upper arms close to the body) is replacing fixed 5 to 20 lb limits. Splint to cough. Risks: diabetes, obesity, smoking.

## ARREST AFTER CARDIAC SURGERY (CSU-ALS)

1. Call arrest: VF, asystole, or flat arterial and ETCO2 traces; resternotomy team gowns up
2. Airway: FiO2 100%, remove PEEP; bag, check tube, decompress suspected tension pneumothorax
3. VF/pVT: up to 3 shocks within 1 minute, before compressions; amiodarone 300 mg if they fail
4. Asystole or extreme brady: pace DDD 80 to 100/min, maximum output; no routine atropine. Paced PEA: pause pacer to unmask VF
5. Not fixed within 1 minute: compressions to arterial systolic over 60; stop infusions (sedation may continue)
6. No routine epinephrine or vasopressin; small doses only if an experienced clinician directs
7. Resternotomy within 5 minutes (first 10 days) or rapid ECMO in expert centers; IABP on pressure trigger

**EXAM TIP** CSU-ALS vs ACLS: shock or pace before compressions; no AEDs or mechanical CPR devices. Systolic under 60 despite good compressions suggests tamponade or bleeding.

Follow your institution's protocols for all doses.

**Sources** (23) are listed with links at icubasics.com/sheets/cardiac-surgery-postop.html. For education and exam preparation only. Not medical advice; follow current guidelines and your institution's protocols. ICU Basics is an independent study aid not affiliated with any certifying body.